

Classical Demand II: Duality and Slutsky

Advanced Microeconomics 1: Economic Decision Making. Lecture 3

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Where we left the question

- **Lecture 2 built the ideal answer:** the correct indexation payment for household h is $e^h(p^1, u^0) - w$.
- **And admitted the catch:** e , h , v are all constructed from u — and no dataset contains u . The answer is ideal and uncomputable.
- **What data do contain:** Marshallian demand $x(p, w)$ — purchases at prices and budgets. Supermarket scanners, the Household Budget Survey, your quiz.

Today: **four bridges from x to the unobservables** — Shephard, the second-derivative structure, the Slutsky equation, Roy. All four are derivative identities. By the end, the size of the indexation error is a formula in estimable objects.

The one idea behind all four bridges

$e(p, u)$ and $v(p, w)$ are **value functions** — the optimized worth of a problem, as its parameters move.

- Differentiating a value function is easier than it looks. When p_ℓ ticks up, two things happen: the old plan costs more (*direct effect*), and the plan re-optimizes (*indirect effect*).
- At an optimum, the indirect effect is **second-order**: the plan was already the best available, so small adjustments to it are worth nothing at the margin.
- So only the direct effect survives — and the direct effect is always something *simple*. This is the envelope logic; the next four theorems are four instances of it.

Bridge 1: the index's slope is a demand

Shephard's lemma

Maintained: u continuous, representing LNS and strictly convex $\succsim$; h single-valued, differentiable where needed.

Theorem · Shephard's lemma (MWG 3.G.1)

For all $p \gg 0$ and u :
$$h(p, u) = \nabla_p e(p, u).$$

▶ proof

- Envelope logic in one line: raise p_ℓ by a cent, and the cost of staying as well off rises by — how much you buy of good ℓ . Substitution is second-order.
- Read it for the thread: **the true cost-of-living index's slope is a demand**. The unobservable e has an observable derivative. First contact between the ideal answer and the data.

Differentiate once more

Theorem · MWG 3.G.2

With $h(\cdot, u)$ continuously differentiable at (p, u) , the $L \times L$ matrix $D_p h(p, u)$ satisfies:

- (i) $D_p h(p, u) = D_p^2 e(p, u)$;
- (ii) $D_p h(p, u)$ is **negative semidefinite** and **symmetric**;
- (iii) $D_p h(p, u) p = 0$.

► proof

- (ii) NSD is the differential compensated law of demand — and it is exactly **the concavity of e** , Lecture 2's sleeper property, now cashed in.
- (ii) **symmetry** is new information: cross-effects of compensated demand are equal, $\partial h_\ell / \partial p_k = \partial h_k / \partial p_\ell$ — Young's theorem applied to e . Hold this thought two slides.

Bridge 2: the meeting point

The Slutsky equation

Theorem · Slutsky equation (MWG 3.G.3)

For all (p, w) and $u = v(p, w)$:

$$\underbrace{\frac{\partial h_\ell(p, u)}{\partial p_k}}_{\text{unobservable}} = \underbrace{\frac{\partial x_\ell(p, w)}{\partial p_k} + \frac{\partial x_\ell(p, w)}{\partial w} x_k(p, w)}_{\text{estimable from demand data}}$$

► proof

- The right side is *exactly* Lecture 1's S — built from choices, no utility anywhere. **The two constructions of the course meet:** $D_p h(p, u) = S(p, w)$.
- “Extremely important empirically” (Sareh’s notes): h is unobservable, yet its derivative matrix is computable from Marshallian data — every applied demand system lives here (Deaton & Muellbauer 1980).

What the meeting point settles

- **Preference maximization** $\Rightarrow$ **S symmetric, NSD, $Sp = 0$** . Lecture 1 got NSD + singularity from WARP alone and flagged symmetry as *conspicuously missing*. Now we see what fills the gap: symmetry is the demand-level fingerprint of **transitivity** — one coherent ranking behind all choices.
- So the preference-based approach imposes *strictly more* structure on data than WARP: the gap between the two routes is exactly one matrix condition. Testable — econometricians test it (and Lecture 4 shows why a rejection would kill the compensation program).
- **Decomposition.** Rearranged: $\frac{\partial x_\ell}{\partial p_k} = \frac{\partial h_\ell}{\partial p_k} - \frac{\partial x_\ell}{\partial w} x_k$ — every price response splits into *substitution* (signed, by the compensated law) and *income* (signed by normality). Giffen goods decoded: $\partial x_\ell / \partial p_\ell > 0$ requires an inferior good whose income effect overwhelms substitution.

Substitutes and complements, finally well-posed

Definition · net substitutes / complements

Goods ℓ and k are **(net) substitutes** if $\partial h_\ell / \partial p_k \geq 0$ and **(net) complements** if < 0 .

- Defined on *Hicksian* demand deliberately: symmetry makes the relation mutual (ℓ substitutes for $k \iff k$ for ℓ), which the Marshallian “gross” notion fails — asymmetric income effects can make ℓ a gross substitute for k but not conversely.
- From $D_p h p = 0$ and $\partial h_\ell / \partial p_\ell \leq 0$: **every good has at least one net substitute**. Substitution is not optional in this theory — which is precisely why fixed baskets cannot be exactly right.
- The IOU from Lecture 1’s comparative statics section: paid.

Bridge 3: from v back to x — and back again

Roy's identity

Theorem · Roy's identity (MWG 3.G.4)

If v is differentiable at $(p, w) \gg 0$, then for every ℓ :

$$x_{\ell}(p, w) = - \frac{\partial v(p, w) / \partial p_{\ell}}{\partial v(p, w) / \partial w}.$$

► proof

- Forward reading (Sareh's): demand from v by differentiation — far easier than solving first-order conditions. The workhorse demand systems (AIDS, QUAIDS) are *born* this way: posit a flexible v , differentiate.
- **Backward reading (ours)**: Roy is a differential equation linking an *estimated* $x(p, w)$ to the unknown v . Solve it, and the data hand you indirect utility — hence e , hence the ideal answer. This is precisely Hausman's (1981) engine, executed in Lecture 4.

The answer, in estimable form

CV and EV are areas under Hicksian demand

One price changes: $p_1^0 \rightarrow p_1^1$, others fixed at p_{-1} . Since $h_1 = \partial e / \partial p_1$ (Shephard), the fundamental theorem of calculus gives

$$-CV = e(p^1, u^0) - e(p^0, u^0) = \int_{p_1^0}^{p_1^1} h_1(t, p_{-1}, u^0) dt, \quad -EV = \int_{p_1^0}^{p_1^1} h_1(t, p_{-1}, u^1) dt.$$

- The ideal answer is **an area under a compensated demand curve** — u^0 -curve for CV, u^1 -curve for EV. They bracket the Marshallian curve for a normal good.
- And the integrand is reachable: the Slutsky equation delivers h 's derivatives from estimable objects, Roy delivers v from an estimated x . **The uncomputable answer of Lecture 2 is now a computation.**

Second-order Taylor expansion of $e(\cdot, u^0)$ around p^0 , using Shephard ($\nabla_p e = h = x^0$ at (p^0, u^0)) and $D_p^2 e = S$:

$$e(p^1, u^0) \approx \underbrace{p^1 \cdot x^0}_{\text{basket payment}} + \underbrace{\frac{1}{2} \Delta p^T S(p^0, w) \Delta p}_{\leq 0 \text{ (NSD)}}$$

- So the **substitution bias** of basket indexation is $\Delta p^T S \Delta p / 2$ in euros — Lecture 1's inequality, now a *formula*, in the matrix Lecture 1 built from choices alone.
- Check against Lecture 2's toy: poor Cobb–Douglas household, energy +50% — exact bias 0.9pp of budget; this quadratic approximates it to first digits. The bigger and the more *relative* the price change, the bigger the bias: 2022 was maximally both.
- Measured in the wild: over the UK's 2021–23 surge, substitution cut the cost of maintaining pre-surge living standards by 1.5pp against a 26.2% Laspeyres rise (Chen, Levell & O'Connell 2025). Lecture 4 uses their full accounting.

Practitioners often report *Marshallian* consumer surplus: $\Delta CS = \int_{p_1^1}^{p_1^0} x_1(t, p_{-1}, w) dt$ — no u anywhere, one estimated curve. When is that the ideal answer rather than a third, wrong number?

Theorem · no wealth effects (Marshall's case)

If $\partial x_1(p, w)/\partial w = 0$ for all (p, w) in the relevant range, then

$$CV = \Delta CS = EV.$$

► proof

- Plausible when the good is a small budget share — and *that is an empirical statement about who you are*: energy is small for the rich household, a tenth of the budget for the poor one. **The shortcut's legality varies across the income distribution** — one more reason one number for everyone misleads.

Where the question stands

Takeaways

- **Four bridges built:** $h = \nabla_p e$ (Shephard); $D_p h = D_p^2 e$ symmetric NSD; $D_p h = S$ (Slutsky) — the course's two constructions are one; x from v (Roy), and v from x read backwards.
- **The answer is estimable:** CV/EV are areas under Hicksian curves; the substitution bias of basket indexation is $\Delta p^T S \Delta p / 2$; the CS shortcut is licensed exactly when wealth effects vanish.
- **One debt outstanding:** all of this assumed a rational household *exists* behind the data. When does observed demand actually pin one down — and what certifies your 25 quiz choices? Lecture 4: integrability, SARP, Afriat.

Required: MWG 3.G; skim 3.H before next time.

Papers: finish Hausman (AER 1981) — Lecture 4 walks its pipeline on the thread.

Ahead: bring your practical output (CCEI in hand) to Lecture 4.

Appendix: proofs

Proof: Shephard's lemma (MWG 3.G.1)

Assume $h(p, u) \gg 0$ single-valued and differentiable. Write $e(p, u) = p \cdot h(p, u)$ and differentiate in p :

$$\nabla_p e(p, u) = h(p, u) + [D_p h(p, u)]^T p.$$

The second term vanishes:

- EMP first-order conditions: $p = \lambda \nabla u(h(p, u))$ for some $\lambda \geq 0$.
- “No excess utility” holds identically in p : $u(h(p, u)) = u$. Differentiate in p :
 $[D_p h]^T \nabla u(h) = 0$.
- Hence $[D_p h]^T p = \lambda [D_p h]^T \nabla u(h) = 0$, and $\nabla_p e = h$. ■

The envelope logic made exact: re-optimization ($D_p h$) is orthogonal to the price vector, so only the direct effect — the bundle itself — remains.

Proof: properties of $D_p h$ (MWG 3.G.2)

- (i) Shephard: $h(p, u) = \nabla_p e(p, u)$ identically. Differentiate in p : $D_p h = D_p^2 e$.
- (ii) $e(\cdot, u)$ is twice continuously differentiable here, so its Hessian is **symmetric** (Young's theorem); and e is **concave** in p (Lecture 2), so its Hessian is **negative semidefinite**. Both properties transfer to $D_p h$ by (i).
- (iii) $h(\cdot, u)$ is homogeneous of degree zero in p (Lecture 2). Differentiate $h(\alpha p, u) = h(p, u)$ in α at $\alpha = 1$ (Euler): $D_p h(p, u) p = 0$. ■

NSD here is the differential form of the compensated law: $dp \cdot dh = dp^T D_p h dp \leq 0$.

Proof: the Slutsky equation (MWG 3.G.3)

Fix (p, w) and let $u = v(p, w)$; duality gives the identity, in p :

$$h_\ell(p, u) = x_\ell(p, e(p, u)).$$

Differentiate both sides with respect to p_k (chain rule on the second argument):

$$\frac{\partial h_\ell(p, u)}{\partial p_k} = \frac{\partial x_\ell(p, e(p, u))}{\partial p_k} + \frac{\partial x_\ell(p, e(p, u))}{\partial w} \cdot \frac{\partial e(p, u)}{\partial p_k}.$$

Two substitutions finish it: by Shephard, $\partial e / \partial p_k = h_k(p, u)$; and at $u = v(p, w)$ we have $e(p, u) = w$ and $h_k(p, u) = x_k(p, w)$. Hence

$$\frac{\partial h_\ell(p, u)}{\partial p_k} = \frac{\partial x_\ell(p, w)}{\partial p_k} + \frac{\partial x_\ell(p, w)}{\partial w} x_k(p, w) = s_{\ell k}(p, w). \blacksquare$$

Proof: Roy's identity (MWG 3.G.4)

Let $\bar{u} = v(\bar{p}, \bar{w})$. Duality gives the identity, in p :

$$v(p, e(p, \bar{u})) = \bar{u}.$$

Differentiate with respect to p_ℓ at $p = \bar{p}$ (chain rule):

$$\frac{\partial v(\bar{p}, \bar{w})}{\partial p_\ell} + \frac{\partial v(\bar{p}, \bar{w})}{\partial w} \cdot \frac{\partial e(\bar{p}, \bar{u})}{\partial p_\ell} = 0.$$

By Shephard, $\partial e / \partial p_\ell = h_\ell(\bar{p}, \bar{u}) = x_\ell(\bar{p}, \bar{w})$ (duality at $\bar{u} = v(\bar{p}, \bar{w})$). Since $\partial v / \partial w > 0$:

$$x_\ell(\bar{p}, \bar{w}) = - \frac{\partial v(\bar{p}, \bar{w}) / \partial p_\ell}{\partial v(\bar{p}, \bar{w}) / \partial w}. \blacksquare$$

Proof: no wealth effects $\Rightarrow CV = \Delta CS = EV$

Suppose $\partial x_1(p, w)/\partial w = 0$ on the relevant range. Use $h_1(p, u) = x_1(p, e(p, u))$ and differentiate in u :

$$\frac{\partial h_1(p, u)}{\partial u} = \frac{\partial x_1}{\partial w} \cdot \frac{\partial e(p, u)}{\partial u} = 0.$$

- So h_1 does not depend on u at all: $h_1(p, u^0) = h_1(p, u^1) = x_1(p, w)$ for every p on the path.
- The three areas are integrals of the *same* function:

$$-CV = \int_{p_1^0}^{p_1^1} h_1(t, u^0) dt = \int_{p_1^0}^{p_1^1} x_1(t, w) dt = \int_{p_1^0}^{p_1^1} h_1(t, u^1) dt = -EV. \blacksquare$$

Quasilinear utility $u = x_1$ -free numeraire form delivers this globally; small budget shares deliver it approximately. Sareh's Exercise 5.3 is this theorem in disguise.